UCE405: SURVEYING

L	T	P	Cr
3	0	3	4.5

Course Objective: Surveying as a subject in civil engineering aims to provide basic knowledge about principles of surveying for a location, and its application in design and construction of engineering projects. The students develop skills using surveying instruments including measuring tapes, theodolites, and advanced measurement equipment such as total stations.

Surveying: Definition, classification of surveys, Principle, distorted or shrunk scales, precision in surveying.

Chain Surveying: Types of chains, tapes, ranging–direct & indirect, chaining on sloping ground, mistakes in chaining, corrections for linear measurements. Reconnaissance, station selection, limiting length of offsets, field notes.

Compass Traversing: Types of compasses, bearings, meridians, declination, dip of magnetic needle, bearing of lines from included angles, local attraction, closing error and its removal.

Theodolite: Types of theodolites, measurement of angles, temporary and permanent adjustments, closed & open traverse, omitted measurements, consecutive and independent co-ordinates, advantages and disadvantages of traversing closing error, Bowditch & Transit Rules

Leveling: Definitions of terms used in leveling, different types of levels, parallax, staff, temporary adjustments, bench marks, booking and reducing the levels, rise and fall method, line of collimation method, errors in leveling, permanent adjustments, corrections to curvature and refraction, setting out grades.

Contours: Definition, representation of reliefs, horizontal equivalent, contour interval, characteristics of contours, methods of contouring, contour gradient, uses of contour maps.
Plane Table Surveying: Introduction to plane table surveying, principle, instruments, working operations, setting up the plane table, centering, leveling, Orientation, methods of plane table survey, two- and three-point problems, danger circle, Lehmann's Rules, errors.

Tacheometry: Definitions and terms used in tacheometry, determination of constants, angular tacheometry with staff vertical and staff inclined, Merits and Demerits; Analytic lens, tangential method of tachometry, subtense method of tacheometry.

Trigonometric Leveling: Definitions & terms, curvature & refraction Methods: direct & reciprocal, eye and object correction, coefficient of refraction.

Curves setting: Definition, elements of a simple curve, different methods of setting out a simple circular curve, elements of a compound curve, reverse curves, transition curves, their characteristics and setting out, vertical curves, setting out vertical curves, sight distances.

Total Station: Working principle and survey with total station.

Global Positioning Systems (GPS): Working principle, Types of GPS, Corrections, Application of GPS. DGPS-working principle.

Digital Elevation model: Introduction and application

Field astronomy: Introduction, basic principle, and application

Remote sensing: Basic concepts, Principle, and applications Photogrammetry: Concepts and application for map preparation

Laboratory Work

1. Measurement of distances/offsets, and bearings with chain and tape, and compass
2. Levelling Exercises.
3. Measurement of vertical and horizontal angles with theodolite.
4. Tacheometric Survey and determination of tacheometric constants.
5. Plane table survey of an area.
6. Setting out curves.
7. Surveying with Total Station.
8. Survey with DGPS

Course Learning Objectives (CLO)

The students will be able to:

1. Measurement of linear distances with various instruments.
2. Measurement of angles with various instruments.
3. Prepare map of an area surveyed
4. Measure topographical features of an area and prepare topographic map.
5. Setting out of curves.

Text Books

1. Anderson and Mikhail. Surveying Theory and Practice, 7th Edition, McGraw Hill Education (2012)
2. Duggal, S.K. Surveying, Vol.I and II, 5th Edition, (2019)
3. Subramanian,R. Surveying and Levelling,Oxford 2nd edition (2012)
4. Venkatramaiah,C.,A Text Book of Surveying, Universities Press(2011)

Reference Books

1. Punmia, B.C., Jain, Ashok Kumar and Jain, Arun Kumar, Surveying Vol.I and II, Laxmi Publications(2005)
2. Agor,R., Surveying, Khanna Publishers (1991)
3. Singh,Narinder, Surveying,Tata McGraw Hill (1992)

Evaluation Scheme

Sr. No.	**Evaluation elements**	**Weightage (%)**
1	MST	25
2	EST	35
3	Sessional: (May include the following) Assignment, Regular Lab assessmentand Quizzes, Project (Including report, presentation etc.)	40